Arithmetic sequences

1

a No **b** Yes **c** No **d** Yes

e No **f** No **g** Yes **h** No

2 By choosing any term after the first and subtracting the preceding term from it.

3

a $d = -2$ **b** $d = 11$ **c** $d = -50$ **d** $d = \frac{3}{7}$

4

a $16, 20,$ **b** $-10, -14$ **c** $24, 29$ **d** $8, 9.5$

5

a $12, 24$ **b** $-5, 5, 15$

6

a $-10, -6, -2, 2$ **b** $-8, -10, -12, -14$

c $a, a + d, a + 2d, a + 3d$

7

a $d = 0.9$ **b** $n = 11$

8 $-6, 0, 6, 12, 18$

9 $x = -7$

10

a $d = 4$ **b** $u_1 = 5$ **c** $u_{11} = 45$

11

a $d = -3$ **b** $u_1 = 4$ **c** -35

12 $d = -12$

13

a $n \leq 5$ **b** 5 **c** 6

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14

a $d = \dfrac{y - x}{4}$ **b** $u_1 = \dfrac{9x - 5y}{4}$ **c** $\dfrac{-3x + 5y}{2}$

15

a $u_1 = 0$ **b** $u_9 = -16$

16 $n = 27$

17 $n = 117$

Explicit rules

18

a

i $d = 7$ **ii** $u_n = 7n - 2$ **iii** $u_{10} = 68$

b

i $d = -0.8$ **ii** $u_n = -0.8n + 17.8$

iii $u_{10} = 9.8$

c

i $d = -7$ **ii** $u_n = -7n + 17$

iii $u_{10} = -53$

d

i $d = \dfrac{3}{4}$ **ii** $u_n = \dfrac{3x + 17}{4}$

iii $u_{10} = \dfrac{47}{4}$

19

a

i $11, 14, 17, 20$ **ii** 3

b

i $11, 21, 31, 41$ **ii** 10

c

i $-7, -10, -13, -16$ **ii** -3

20

a

i $u_1 = 15$ **ii** $d = 5$ **iii** $u_5 = 35$

b

i $u_1 = 20$ **ii** $d = -8$ **iii** $u_5 = -12$

c

i $u_1 = 25$ **ii** $d = -4$ **iii** $u_5 = 9$

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d

i $u_1 = 3$ ii $d = 12$ iii $u_5 = 51$

21

a 6 **b** 8 **c** $u_n = 6n + 2$ **d** 152

Applications

22 $2670

23 1510 cm

24

a 50 m **b** 200 m **c** 500 m

25

a $d = 35$ **b** $u_1 = 151$ **c** $n = 22$

26

a $1\,800$ **b** $V_n = 1\,800 - 50n$

c $n = 18$

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